15.095 Machine Learning HW2

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Question 1

(a)

False.

Consider the underlying function

$$f(x) = \sin x,$$

which is non-convex on $\mathbb{R}$. Let $n = 2$ and choose

$$x_1 = 0, \qquad x_2 = \frac{\pi}{2}.$$

Then

$$y_1 = f(x_1) = 0, \qquad y_2 = f(x_2) = 1.$$

So we can set:

$$\theta_1 = 0, \qquad \theta_2 = 1, \qquad \xi_1 = \xi_2 = \frac{1-0}{\frac{\pi}{2}-0} = \frac{2}{\pi}.$$

The convexity constraints are satisfied since

$$\theta_1 + \xi_1(x_2 - x_1) = 0 + \frac{2}{\pi}\left(\frac{\pi}{2}\right) = 1 = \theta_2,$$

and

$$\theta_2 + \xi_2(x_1 - x_2) = 1 + \frac{2}{\pi}\left(-\frac{\pi}{2}\right) = 0 = \theta_1.$$

Moreover, since $\theta_i = y_i$ for both observations,

$$\frac{1}{2}\sum_{i=1}^{2}(y_i - \theta_i)^2 = 0.$$

Thus, even though the underlying function f is non-convex, the convex regression problem only needs to fit the observed points. These two sampled points lie on the affine function $g(x) = \frac{2}{\pi}x$, which is convex. Hence, they can be fit perfectly, giving an objective value of zero. Therefore, the statement is false. □

(b)

True.

Let

$$r_i = y_i - x_i^\top \beta, \qquad i = 1, \ldots, n.$$

Introduce binary variables $z_i \in \{0, 1\}$, where $z_i = 1$ means that residual i is selected. We also introduce continuous variables $u_i \geq 0$. We can formulate the problem as

$$\min_{\beta, r, u, z} \sum_{i=1}^{n} u_i$$

subject to

$$r_i = y_i - x_i^\top \beta, \qquad i = 1, \ldots, n,$$

$$u_i \geq r_i - M(1 - z_i), \qquad i = 1, \ldots, n,$$

$$u_i \geq -r_i - M(1 - z_i), \qquad i = 1, \ldots, n,$$

$$\sum_{i=1}^{n} z_i = k,$$

$$u_i \geq 0, \qquad i = 1, \ldots, n,$$

$$z_i \in \{0, 1\}, \qquad i = 1, \ldots, n,$$

where M is a sufficiently large constant.

If $z_i = 1$, the two Big-M constraints reduce to

$$u_i \geq r_i, \qquad u_i \geq -r_i,$$

which is equivalent to

$$u_i \geq |r_i|.$$

If $z_i = 0$, the Big-M constraints become

$$u_i \geq r_i - M, \qquad u_i \geq -r_i - M.$$

For sufficiently large M, both right-hand sides are less than or equal to zero. Thus, together with $u_i \geq 0$, the smallest feasible value is $u_i = 0$. Since the objective minimizes $\sum_{i=1}^{n} u_i$, the optimizer will set $u_i = 0$ for these unselected residuals.

Since $\sum_{i=1}^{n} z_i = k$, exactly k residuals must be selected.

For each selected residual, $u_i \geq |r_i|$, and since we minimize $\sum_{i=1}^{n} u_i$, we have $u_i = |r_i|$ at optimality.

For each unselected residual, $u_i = 0$. Therefore, minimizing $\sum_{i=1}^{n} u_i$ selects and sums the k smallest absolute residuals. All constraints and the objective are linear, with z_i binary, so this is a linear MIO formulation. □

(c)

False.

The two constraints do different things. For two highly correlated variables, the constraint

$$z_i + z_j \leq 1$$

means that we can select at most one of them. So the possible choices are

$$(z_i, z_j) \in \{(0,0), (1,0), (0,1)\}.$$

On the other hand, if we put i and j in the same group and impose

$$z_i = z_j,$$

then they must either both be selected or both be excluded. The possible choices are therefore

$$(z_i, z_j) \in \{(0,0), (1,1)\}.$$

These are not equivalent. In fact, $z_i = z_j$ allows both highly correlated variables to be selected at the same time, while $z_i + z_j \leq 1$ is specifically used to prevent this. Therefore, we cannot replace the pairwise-collinearity constraint with the group-sparsity constraint. □

(d)

False.

From the robust logistic regression result in Lecture 6, if the feature uncertainty set is

$$\mathcal{U}_x = \{\Delta X : \|\Delta x_i\|_q \leq \rho, \quad i = 1, \ldots, n\},$$

then the robust counterpart is

$$\max_{\beta, \beta_0} -\sum_{i=1}^{n} \log\left(1 + e^{-y_i(\beta^\top x_i + \beta_0) + \rho\|\beta\|_{q^*}}\right),$$

where q^* is the dual norm of q, satisfying

$$\frac{1}{q} + \frac{1}{q^*} = 1.$$

Thus, the robustness term is generally

$$\rho\|\beta\|_{q^*},$$

not

$$\rho\|\beta\|_q.$$

For example, take

$$q = \infty, \qquad \rho = 1, \qquad \beta = (1,1).$$

The dual norm of ℓ_∞ is ℓ_1, so the correct robustness term is

$$\rho\|\beta\|_1 = 1(1+1) = 2.$$

However, the expression in the statement would use

$$\rho\|\beta\|_\infty = 1 \cdot 1 = 1.$$

Therefore, the stated robust counterpart underestimates the worst-case effect and is not correct in general. The two expressions coincide only for self-dual norms, such as $q = 2$. □

(e)

True.

Let

$$m_i = y_i(\beta^\top x_i + \beta_0).$$

The robust problem:

$$\max_{\beta,\beta_0} \min_{\Delta y \in \mathcal{U}_y} -\sum_{i=1}^{n} \log\left(1 + e^{-(1-2\Delta y_i)m_i}\right),$$

where

$$\mathcal{U}_y = \left\{ \Delta y \in \{0,1\}^n : \sum_{i=1}^{n} \Delta y_i \le \Gamma \right\}.$$

For fixed β, β_0, we define

$$f_i(\Delta y_i) = -\log\left(1 + e^{-(1-2\Delta y_i)m_i}\right).$$

Since $\Delta y_i \in \{0,1\}$, we can write

$$f_i(\Delta y_i) = f_i(0) + \big(f_i(1) - f_i(0)\big)\Delta y_i.$$

$$f_i(0) = -\log(1 + e^{-m_i}),$$

$$f_i(1) - f_i(0) = \log\left(\frac{1 + e^{-m_i}}{1 + e^{m_i}}\right).$$

Thus the inner problem becomes

$$\min_{\Delta y} -\sum_{i=1}^{n} \log(1 + e^{-m_i}) + \sum_{i=1}^{n} c_i \Delta y_i,$$

where

$$c_i = \log\left(\frac{1 + e^{-m_i}}{1 + e^{m_i}}\right),$$

subject to

$$\Delta y_i \in \{0,1\}, \qquad \sum_{i=1}^{n} \Delta y_i \le \Gamma.$$

Now we observe that

$$0 \le \Delta y_i \le 1, \qquad \sum_{i=1}^{n} \Delta y_i \le \Gamma$$

has integer extreme points when Γ is integer, we can therefore relax the binary restriction, which would not change the optimal value. So the inner problem is equivalent to the LP

$$\min_{\Delta y} \sum_{i=1}^{n} c_i \Delta y_i$$

subject to

$$\sum_{i=1}^{n} \Delta y_i \le \Gamma,$$
$$\Delta y_i \le 1, \qquad i = 1, \dots, n,$$
$$\Delta y_i \ge 0, \qquad i = 1, \dots, n.$$

Now we can take the dual problem. By strong duality, the optimal value is equal to that of its dual problem.

Let $\mu \le 0$ be the dual variable associated with

$$\sum_{i=1}^{n} \Delta y_i \le \Gamma,$$

and let $\nu_i \leq 0$ be the dual variable associated with

$$\Delta y_i \leq 1.$$

The dual is

$$\max_{\mu,\nu} \Gamma\mu + \sum_{i=1}^{n} \nu_i$$

subject to

$$\begin{aligned} \mu + \nu_i &\leq c_i, \qquad i = 1, \ldots, n, \\ \mu &\leq 0, \qquad \nu_i \leq 0. \end{aligned}$$

By strong duality, the robust problem is therefore equivalent to

$$\max_{\beta,\beta_0,\mu,\nu} -\sum_{i=1}^{n} \log\left(1 + e^{-y_i(\beta^\top x_i + \beta_0)}\right) + \Gamma\mu + \sum_{i=1}^{n} \nu_i$$

subject to

$$\begin{aligned} \mu + \nu_i &\leq \log\left(\frac{1 + e^{-y_i(\beta^\top x_i + \beta_0)}}{1 + e^{y_i(\beta^\top x_i + \beta_0)}}\right), \qquad i = 1, \ldots, n, \\ \mu &\leq 0, \qquad \nu_i \leq 0. \end{aligned}$$

Finally,

$$\log\left(\frac{1 + e^{-m_i}}{1 + e^{m_i}}\right) = -m_i,$$

so the constraints may equivalently be written as

$$\mu + \nu_i \leq -y_i(\beta^\top x_i + \beta_0).$$

The robust counterpart is therefore a single continuous concave maximization problem, equivalently a convex optimization problem, and contains no integer variables. Therefore the statement is true. □

Question 2

(a)

Fix (β, β_0), so the losses c_i are constants, and we order them such that

$$c_{(1)} \geq c_{(2)} \geq \cdots \geq c_{(n)}.$$

For $z \in \mathcal{Z}$, exactly k entries of z equal 1. Therefore, the objective is maximized by selecting the k largest losses. Hence, it is equivalent to saying

$$\max_{z \in \mathcal{Z}} \sum_{i=1}^{n} z_i c_i = \sum_{j=1}^{k} c_{(j)}.$$

Now consider the continuous relaxation

$$\mathcal{P} = \left\{ z \in \mathbb{R}^n : \sum_{i=1}^{n} z_i = k,\ 0 \leq z_i \leq 1 \right\}.$$

Suppose $z \in \mathcal{P}$ is optimal. However, there exists indices a, b such that

$$c_a > c_b, \qquad z_a < 1, \qquad z_b > 0.$$

That is, some positive weight is placed on a smaller loss c_b while the larger loss c_a has not yet received its maximum weight of 1.

We define

$$\epsilon = \min\{1 - z_a, z_b\} > 0,$$

essentially satisfying the conditions that z_a cannot exceed 1, and z_b cannot go below 0.

We then construct a new solution z' by moving ϵ units of weight from b to a:

$$z'_a = z_a + \epsilon, \qquad z'_b = z_b - \epsilon,$$

with all other components unchanged.

Since the same amount of weight is added and removed,

$$\sum_{i=1}^{n} z'_i = \sum_{i=1}^{n} z_i = k.$$

Also, with our choice of ϵ,

$$0 \leq z'_i \leq 1.$$

Therefore, $z' \in \mathcal{P}$.

Our change in the objective is

$$\sum_{i=1}^{n} z'_i c_i - \sum_{i=1}^{n} z_i c_i = (z_a + \epsilon)c_a + (z_b - \epsilon)c_b - z_a c_a - z_b c_b = \epsilon c_a - \epsilon c_b = \epsilon(c_a - c_b).$$

Since $\epsilon > 0$ and $c_a > c_b$,

$$\epsilon(c_a - c_b) > 0.$$

Thus, moving weight from the smaller loss to the larger loss strictly increases the objective, contradicting our initial assumption that z was optimal.

Therefore, at an optimal point, no positive weight can be placed on a smaller loss while a larger loss has weight less than 1. So there exists an optimal solution that assigns weight 1 to the k largest losses and weight 0 to all remaining losses. In other words,

$$\max_{z \in \mathcal{P}} \sum_{i=1}^{n} z_i c_i = \sum_{j=1}^{k} c_{(j)}.$$

We combine the two results,

$$\max_{z\in\mathcal{Z}} \sum_{i=1}^{n} z_i c_i = \max_{z\in\mathcal{P}} \sum_{i=1}^{n} z_i c_i = \sum_{j=1}^{k} c_{(j)}.$$

□

(b)

By part (a), for fixed (β, β_0), we can replace the inner maximization over $\mathcal{Z}$ by its continuous relaxation $\mathcal{P}$. Let

$$c_i = \ell(y_i, \beta^\top x_i + \beta_0).$$

Then the inner problem is

$$\begin{aligned} \max_{z} \quad & \sum_{i=1}^{n} c_i z_i \\ \text{s.t.} \quad & \sum_{i=1}^{n} z_i = k, \\ & z_i \le 1, \qquad i = 1, \dots, n, \\ & z_i \ge 0, \qquad i = 1, \dots, n. \end{aligned}$$

Associate the free dual variable θ with the constraint (due to the equality, the dual variable is free)

$$\sum_{i=1}^{n} z_i = k,$$

and the dual variable $u_i \ge 0$ with each constraint $z_i \le 1$.

Since each primal variable $z_i \ge 0$, the corresponding dual constraint is

$$\theta + u_i \ge c_i, \qquad i = 1, \dots, n.$$

Therefore, the dual of the inner problem is

$$\begin{aligned} \min_{\theta, u} \quad & k\theta + \sum_{i=1}^{n} u_i \\ \text{s.t.} \quad & \theta + u_i \ge c_i, \qquad i = 1, \dots, n, \\ & u_i \ge 0, \qquad i = 1, \dots, n, \\ & \theta \in \mathbb{R}. \end{aligned}$$

By strong duality, the inner maximization and its dual have the same optimal value. Substituting

$$c_i = \ell(y_i, \beta^\top x_i + \beta_0)$$

and combining the dual with the outer minimization over (β, β_0) gives

$$\boxed{\begin{aligned} \min_{\beta, \beta_0, \theta, u} \quad & k\theta + \sum_{i=1}^{n} u_i \\ \text{s.t.} \quad & \theta + u_i \ge \ell(y_i, \beta^\top x_i + \beta_0), \qquad i = 1, \dots, n, \\ & u_i \ge 0, \qquad i = 1, \dots, n, \\ & \theta \in \mathbb{R}. \end{aligned}}$$

Thus, problem (2.1) is equivalent to formulation (2.2). □

Now let the loss be the misclassification loss $\mathbf{1}\left\{y_i(\beta^\top x_i + \beta_0) \le 0\right\}$.

Here, the indicator function equals 1 when its condition is true and 0 otherwise. Thus,

$$y_i(\beta^\top x_i + \beta_0) \le 0$$

means that observation i is misclassified, so its loss is 1. If

$$y_i(\beta^\top x_i + \beta_0) > 0,$$

the observation is correctly classified and its loss is 0.

Since we first define $c_i = \ell(y_i, \beta^\top x_i + \beta_0)$ and the duality argument holds for any fixed values of c_i, it remains valid when ℓ is the misclassification loss. Therefore, **formulation (2.2) is still valid** as

$$c_i = \mathbf{1}\left\{y_i(\beta^\top x_i + \beta_0) \le 0\right\}.$$

However, **(2.2) cannot be handed directly** to an LP or convex optimization solver. In particular, its constraints become

$$\theta + u_i \ge \mathbf{1}\left\{y_i(\beta^\top x_i + \beta_0) \le 0\right\}.$$

The indicator function jumps from 0 to 1 when $y_i(\beta^\top x_i + \beta_0)$ crosses zero, so it is a discontinuous and nonconvex function of (β, β_0). Therefore, the resulting problem is not directly an LP or convex optimization problem. We might need some further mixed-integer reformulation, for example we can use binary variables and big-M constraints, before we give it to a solver.

(c)

For the hinge loss,

$$\ell(y_i, \beta^\top x_i + \beta_0) = \max\left\{1 - y_i(\beta^\top x_i + \beta_0),\, 0\right\}.$$

Substituting this into (2.2), we have

$$\theta + u_i \ge \max\left\{1 - y_i(\beta^\top x_i + \beta_0),\, 0\right\}.$$

We see how a constraint in the form $a \ge \max\{b, 0\}$ is equivalent to

$$a \ge b \qquad \text{and} \qquad a \ge 0.$$

Therefore, the hinge-loss constraint can be written as

$$\theta + u_i \ge 1 - y_i(\beta^\top x_i + \beta_0)$$

and

$$\theta + u_i \ge 0.$$

Rearrange the first constraint:

$$\theta + u_i + y_i x_i^\top \beta + y_i \beta_0 \ge 1.$$

Thus, (2.2) becomes

$$\begin{aligned}
\min_{\beta,\beta_0,\theta,u} \quad & k\theta + \sum_{i=1}^{n} u_i \\
\text{s.t.} \quad & \theta + u_i + y_i x_i^\top \beta + y_i \beta_0 \ge 1, \qquad i = 1, \dots, n, \\
& \theta + u_i \ge 0, \qquad i = 1, \dots, n, \\
& u_i \ge 0, \qquad i = 1, \dots, n.
\end{aligned}$$

The objective and all constraints are linear, so this is a linear optimization problem (LP).

There are p variables in β, one variable β_0, one variable θ, and n variables u_i. Therefore, there are

$$\boxed{p+n+2}$$

continuous variables.

There are n constraints of the first type, n constraints of the second type, and n nonnegativity constraints on u_i, giving

$$\boxed{3n}$$

constraints in total. Equivalently, there are $2n$ general linear constraints and n nonnegativity bounds.

Now suppose instead that the logistic loss is used:

$$\ell(y_i, \beta^\top x_i + \beta_0) = \log\left(1 + e^{-y_i(\beta^\top x_i + \beta_0)}\right).$$

Then (2.2) contains constraints of the form

$$\theta + u_i \geq \log\left(1 + e^{-y_i(\beta^\top x_i + \beta_0)}\right).$$

Unlike hinge loss, logistic loss is a **smooth nonlinear convex function**, rather than a piecewise-linear function. Therefore, it cannot be represented exactly as a linear mixed-integer optimization problem using only finitely many binary variables and linear constraints.

However, binary variables are not necessary. As in Lecture 6 (and in Problem 1 part (e)), applying the relaxation and strong-duality argument leads to a continuous convex nonlinear optimization problem. Thus, although the formulation is not an LP or an exact linear MIO, it is a convex nonlinear optimization problem and is computationally tractable.

(d)

(i)

At a solution $(\beta^t, \beta_0^t, t^t)$, we define the current losses as

$$c_i^t = \ell\left(y_i, (\beta^t)^\top x_i + \beta_0^t\right), \qquad i = 1, \ldots, n.$$

The separation problem is therefore

$$\max_{z \in \mathcal{Z}} \sum_{i=1}^n z_i c_i^t.$$

By part (a), this problem has a closed-form solution: the optimal z selects the k observations with the largest losses. If

$$c_{(1)}^t \geq c_{(2)}^t \geq \cdots \geq c_{(n)}^t,$$

then the optimal value of the separation problem is

$$\sum_{j=1}^k c_{(j)}^t.$$

So if we have

$$\sum_{j=1}^k c_{(j)}^t > t^t,$$

our corresponding z^* gives a violated constraint, so we add the cut

$$t \geq \sum_{i=1}^n z_i^* \ell(y_i, \beta^\top x_i + \beta_0)$$

and re-solve the restricted problem. Otherwise, no constraint is violated and the current solution is optimal.

Computing the n losses requires $O(np)$ time, since evaluating each $\beta^\top x_i$ requires $O(p)$ operations. Sorting the losses to find the k largest requires $O(n \log n)$ time. Thus, a separation step can be performed in

$$\boxed{O(np + n \log n)}$$

time.

(ii)

One reason to prefer the cutting-plane approach is scalability. Formulation (2.2) introduces n additional variables u_i and at least n additional constraints, so its size grows directly with the number of observations.

However, our cutting-plane master problem has only

$$p + 2$$

variables, which are the p components of β, β_0, and t. It starts with only a small subset of the constraints in (2.3) and adds a new constraint only when the separation problem finds a violated one. Therefore, if relatively few cuts are needed, the optimization problems solved at each iteration can be much smaller than solving (2.2) directly.

Question 3

Part (a)

(i)

Imposing the four candidate constraint sets on every adjacent pair gave the following results:

Candidate	Constraints imposed	Objective	Broken constraints
Forward	39	1.0528	574
Backward	39	1.4079	506
Both	78	2.4454	0
Stepped backwards	39	1.0528	908

Thus, imposing **both the forward and backward constraints** on every adjacent pair is the only candidate that leaves none of the full 1560 convexity constraints violated. Since there are $n - 1 = 39$ adjacent pairs and two constraints are imposed on each pair, this formulation uses $2(n - 1) = 78$ constraints.

Its objective value is 2.4454, which is exactly the same as the objective obtained using all 1560 constraints. Thus, for this one-dimensional problem, the 78 adjacent-pair constraints are sufficient.

(ii)

For the random control sets, increasing the number of constraints generally moved the solution closer to the fully convex fit. The results were

Random constraints	Objective	Broken constraints
39	0.5533	608
78	0.7590	517
156	1.3073	380
312	1.6723	248
624	2.2773	70

As more random constraints are added, the objective gets closer to the full-problem objective of 2.4454, and fewer constraints are violated. However, having more constraints does not necessarily mean they are the right ones. For example, 78 random constraints still leave 517 violations, while the 78 bidirectional adjacent-pair constraints from part (i) leave no violations.

This is because the adjacent-pair constraints directly enforce the ordering of slopes needed for convexity. Random constraints can miss these important neighboring relationships. So we should choose the right constraints instead of choosing more of them.

(iii)

From part (i), we impose both the forward and backward constraints on every adjacent pair $(k, k+1)$.

(1)

We impose on a single adjacent pair $(k, k+1)$. The forward constraint is

$$\theta_k + \xi_k(x_{k+1} - x_k) \leq \theta_{k+1}.$$

Since

$$h_k = x_{k+1} - x_k > 0,$$

we can rewrite this as

$$\xi_k h_k \leq \theta_{k+1} - \theta_k.$$

$$\xi_k \leq \frac{\theta_{k+1} - \theta_k}{h_k} = c_k.$$

The backward constraint is

$$\theta_{k+1} + \xi_{k+1}(x_k - x_{k+1}) \leq \theta_k.$$

Since $x_k - x_{k+1} = -h_k$, this becomes

$$\theta_{k+1} - \xi_{k+1} h_k \leq \theta_k.$$

We rearrange and get:

$$c_k \leq \xi_{k+1}.$$

$$\xi_k \leq c_k \leq \xi_{k+1}.$$

Neither inequality alone is sufficient. The forward constraint only gives $\boxed{\xi_k \leq c_k}$, while the backward constraint only gives $\boxed{c_k \leq \xi_{k+1}}$. Both are needed to connect the slopes on the two sides.

(2)

Now we apply this to every adjacent pair.

The result above holds for every $k = 1, \ldots, n-1$. Therefore, the inequalities can be placed end to end:

$$\xi_1 \leq c_1 \leq \xi_2 \leq c_2 \leq \cdots \leq \xi_{n-1} \leq c_{n-1} \leq \xi_n.$$

Here, we see how both the subgradients and the chord slopes are nondecreasing, i.e.:

$$\boxed{\xi_1 \leq \xi_2 \leq \cdots \leq \xi_n,}$$

and

$$\boxed{c_1 \leq c_2 \leq \cdots \leq c_{n-1}.}$$

This is where the ordering $x_1 < x_2 < \cdots < x_n$ is important. Since $x_{k+1} > x_k$, we have $h_k = x_{k+1} - x_k > 0$, so dividing by h_k does not reverse the direction of the inequalities. Also, we see how the adjacent pairs are $(1,2), (2,3), \ldots, (n-1, n)$, so these consecutive inequalities share a common subgradient and can be linked together in one ordered sequence (like we see above).

(3)

Now we consider two nonadjacent points.

First suppose $i < j$. We want to show that the original convexity constraint

$$\theta_i + \xi_i(x_j - x_i) \leq \theta_j$$

holds even though it was not explicitly imposed.

We are given the height difference and distance telescope are

$$\theta_j - \theta_i = \sum_{k=i}^{j-1} c_k h_k,$$

$$x_j - x_i = \sum_{k=i}^{j-1} h_k.$$

From the ordering in part (2), for every $k = i, \ldots, j-1$,

$$\xi_i \le c_k.$$

Since $h_k > 0$,

$$\xi_i h_k \le c_k h_k.$$

We sum these inequalities:

$$\xi_i \sum_{k=i}^{j-1} h_k \le \sum_{k=i}^{j-1} c_k h_k.$$

Now we use the telescoping expressions:

$$\xi_i (x_j - x_i) \le \theta_j - \theta_i.$$

Therefore,

$$\theta_i + \xi_i (x_j - x_i) \le \theta_j.$$

Now suppose $i > j$. We must also prove the reverse-direction constraint

$$\theta_i + \xi_i (x_j - x_i) \le \theta_j.$$

As before, we use the telescoping expressions,

$$\theta_i - \theta_j = \sum_{k=j}^{i-1} c_k h_k, \qquad x_i - x_j = \sum_{k=j}^{i-1} h_k.$$

The difference from the $i < j$ case is that the ordering from part (2) now gives us

$$c_k \le \xi_i, \qquad k = j, \ldots, i-1.$$

Since $h_k > 0$, when we sum over these intervals we get

$$\theta_i - \theta_j = \sum_{k=j}^{i-1} c_k h_k \le \xi_i \sum_{k=j}^{i-1} h_k = \xi_i (x_i - x_j).$$

We just rearrange and get the result:

$$\theta_i + \xi_i (x_j - x_i) \le \theta_j.$$

Thus, the $i < j$ case proves the constraints from a point to every point to its right, while the $i > j$ case proves the constraints to every point to its left. Since the two cases cover both directions between any pair of points, we have shown that every constraint in the full formulation is automatically satisfied. Therefore, we only need the $2(n-1)$ bidirectional adjacent-pair constraints instead of explicitly imposing all $n(n-1)$ constraints. □

Part (b)

(i)

The worst violation is caused by the supporting plane at point 28, which passes above point 12. The size of the violation is approximately 0.518.

(ii)

The step that fails is **Step (2)** of the proof in part (a)(iii). In one dimension, the points can be ordered as

$$x_1 < x_2 < \cdots < x_n,$$

and the adjacent-pair constraints give

$$\xi_k \leq c_k \leq \xi_{k+1}.$$

These inequalities can then be linked together into the single chain

$$\xi_1 \leq c_1 \leq \xi_2 \leq c_2 \leq \cdots \leq \xi_n.$$

In $\mathbb{R}^2$, however, there is no single left-to-right ordering of the points. The Delaunay neighbours form a two-dimensional mesh rather than a chain.

Also, in one dimension each ξ_i is just one number, so it makes sense to compare two slopes and write something like

$$\xi_i \leq \xi_j.$$

In two dimensions, however, each ξ_i has two components, so there is no single left-to-right slope ordering that works in the same way.

Because of this, the inequalities from neighboring points cannot be linked together like they were in part (a). As a result, checking all Delaunay neighbors in both directions is not enough to guarantee the constraints between points that are farther apart.

Part (c)

(i)

Starting with both directions, it took 2 **rounds** to reach no broken constraints. The final model had 231 **constraints** and an objective value of 2.58541.

Starting with only one direction per edge, it took 6 **rounds** to reach no broken constraints. The final model had 207 **constraints** and the same objective value of 2.58541.

So, starting with both directions took fewer rounds since we started with more constraints already included. Starting with one direction took longer, but it ended up using fewer constraints overall. In the end, both starting points reached the same objective value.

(ii)

If $n = 10{,}000$, the full formulation would contain

$$n(n-1) = 10{,}000(9{,}999) = 99{,}990{,}000$$

constraints.

This is almost 100 million constraints, so solving the full model directly would be much more difficult. The purpose of constraint generation is to avoid including all of these constraints from the start. Instead, we begin with a smaller set and only add constraints that are actually violated.